Pharmacovigilance of Doxorubicin: Mechanism of Action, Adverse Drug Reactions, and Risk-Mitigation Strategies

1. Introduction

Generic name: Doxorubicin
Drug class: Anthracycline antineoplastic antibiotic
Main therapeutic purpose: Treatment of selected cancers, depending on cancer type and treatment regimen
Common nickname: "Red Devil" referring to its red color and potential for serious toxicity; this is an informal term, not an official pharmacological name.

2. Mechanism of action

Doxorubicin exerts its anticancer effects through three major mechanisms:First, it intercalates between DNA base pairs, disrupting DNA replication and transcription. Second, it stabilizes the DNA–topoisomerase II complex and prevents re-ligation of DNA strand breaks, producing persistent DNA damage and apoptosis. Third, redox cycling generates reactive oxygen species (ROS), which damage cellular membranes, proteins, mitochondria and DNA. These actions kill tumour cells but can also injure rapidly dividing normal cells. Oxidative stress, mitochondrial dysfunction and topoisomerase IIβ-related injury are especially relevant to cardiac toxicity [2,3,5].

3. Important adverse drug reactions

3.1. Cardiotoxicity: Doxorubicin may cause reduced left ventricular ejection fraction (LVEF), cardiomyopathy, arrhythmias and congestive heart failure. Proposed mechanisms include ROS-mediated oxidative stress, mitochondrial dysfunction, lipid peroxidation, DNA injury involving topoisomerase IIβ and disturbed calcium handling in cardiomyocytes [2,5]. This reaction can be severe and sometimes irreversible, and risk generally rises with cumulative exposure and other cardiotoxic treatments [3,5]. Symptoms may include breathlessness, fatigue, reduced exercise tolerance, leg swelling, rapid weight gain, palpitations or difficulty lying flat, although cardiac dysfunction may initially be asymptomatic. Assess cardiac history and baseline LVEF by echocardiography or MUGA scan, repeat assessment according to treatment and risk, and monitor symptoms, vital signs and cumulative dose; abnormal LVEF or clinical heart-failure findings are important evidence of toxicity [3,5].

3.2. Myelosuppression: Doxorubicin damages rapidly dividing bone-marrow precursor cells, reducing production of neutrophils, red cells and platelets [2,3]. Neutropenia may lead to serious infection, febrile neutropenia or sepsis; anemia can cause marked fatigue and breathlessness, while thrombocytopenia can result in bleeding. Warning signs include fever,

chills, sore throat, pallor, dizziness, easy bruising, petechiae or gum/nose bleeding. A complete blood count (CBC) with differential should be checked before each cycle and monitored around the expected nadir; the prescribing information notes that neutrophil nadir commonly occurs around days 10–14, with recovery often by day 21. CBC results, fever or infection findings guide urgent assessment, supportive care and protocol-based dose delay or adjustment [3].

3.3. Mucositis/stomatitis: Injury to rapidly dividing oral and gastrointestinal epithelial cells, together with oxidative stress, can weaken the mucosal barrier and cause inflammation, redness and ulceration [2,3]. Severity ranges from discomfort to painful ulceration that limits eating and drinking, causing dehydration, weight loss, infection or interruption of chemotherapy. Patients may report mouth pain, burning, ulcers, difficulty swallowing, abdominal discomfort or diarrhea. Inspect the mouth at visits and ask about pain, swallowing, oral intake and bowel symptoms; visible lesions, reduced intake and dehydration are clinical findings to document. Oral hygiene, hydration, appropriate mouth rinses, pain control, nutritional support and prompt treatment of secondary infection can reduce complications [3].

3.4. Extravasation and tissue injury: Doxorubicin is a vesicant, and leakage from the vein into surrounding tissue during intravenous administration can produce direct cellular injury, inflammation and tissue necrosis [3,6]. Severity may range from local pain and swelling to blistering, ulceration, deep necrosis and damage requiring surgical treatment. Early signs include burning, stinging, pain, redness, swelling or altered infusion flow; later findings can include blisters, ulceration and dark discoloration. Check the IV site before and throughout administration. If leakage is suspected, stop the infusion immediately and follow the institutional extravasation protocol; management may include aspiration where appropriate, cold application and prompt consideration of dexrazoxane according to the product guidance. Site findings and timely clinical assessment are the key evidence for detection [3,6].

3.5. Nausea and vomiting: Doxorubicin can provoke nausea and vomiting through effects on gastrointestinal pathways and emetic signaling [3]. These symptoms may range from transient discomfort to persistent vomiting with dehydration, electrolyte disturbance and inability to maintain oral intake. Patients may report nausea, retching, vomiting, poor appetite or reduced fluid intake. Assess symptoms, frequency of vomiting, hydration and oral intake during and after treatment; fluid status and electrolytes may need evaluation when symptoms are substantial. Use guideline-based antiemetic prophylaxis and supportive hydration, and promptly reassess uncontrolled symptoms [3].

3.6. Alopecia: Doxorubicin affects rapidly dividing hair-follicle cells, which can cause partial or complete hair loss [2,3]. Although generally not medically dangerous and often reversible after treatment, alopecia can cause significant distress and affect quality of life. Patients may notice increased shedding or loss of scalp and body hair. Assess and discuss the expected pattern and psychosocial impact before treatment, provide practical support and explain that hair commonly regrows after chemotherapy ends, although timing and texture can vary [3].

4. Measures to Reduce the Risk of Adverse Reactions

4.1. Cardiotoxicity: Review the patient's cardiac history, cumulative anthracycline exposure, and concurrent cardiotoxic medicines. Perform baseline and risk-appropriate follow-up cardiac assessments.
4.2. Myelosuppression: Check the complete blood count (CBC) before each treatment cycle and educate patients to report fever immediately.
4.3. Mucositis: Encourage good oral hygiene, adequate hydration, and proper nutrition. Assess oral lesions early to prevent complications.
4.4. Extravasation: Ensure secure IV access and administration by trained staff. Monitor the injection site continuously, stop the infusion immediately if extravasation is suspected, and follow the local management protocol.
4.5. Nausea and vomiting: Provide appropriate antiemetic prophylaxis and advise patients to report persistent vomiting promptly.
4.6. Other serious symptoms: Educate patients to report breathlessness, chest symptoms, leg swelling, unusual bleeding, and injection-site pain or swelling promptly. [3,5,6]

References

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